$$2SO_{2(g)} + O_{2(g)} \xrightarrow{Pt_{(s)} \text{ or } V_2O_{5(s)}} 2SO_{3(g)}$$

ii) In the Haber's process for the manufacture of ammonia, iron is used as a catalyst for the reaction between Hydrogen and Nitrogen.

$$N_{2(g)} + 3H_{2(g)} \xrightarrow{Fe_{(s)}} 2NH_{3(g)}$$

iii) Oxidation of ammonia is carried out in presence of platinum gauze

$$4NH_{3(g)} + 5O_{2(g)} \xrightarrow{Pt_{(s)}} 4NO_{(g)} + 6H_2O_{(g)}$$

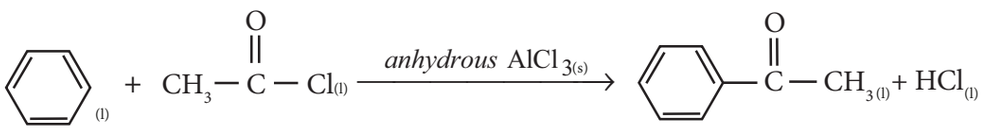
iv) The hydrogenation of unsaturated organic compounds is carried out using finely divided nickel as a catalyst.

$$CH_2{=}CH_{2(g)} + H_{2(g)} \xrightarrow{Ni_{(s)}} CH_3\text{-}CH_{3(g)}$$

v) Decomposition of H_2O_2 occurs in the presence of the Pt catalyst

$$2H_2O_{2(l)} \xrightarrow{Pt_{(s)}} 2H_2O_{(l)} + O_{2(g)}$$

vi) In the presence of anhydrous $AlCl_3$, benzene reacts with ethanoyl chloride to produce acetophenone



acetophenone

10.2.1 Characteristics of catalysts

1. For a chemical reaction, catalyst is needed in very small quantity. Generally, a pinch of catalyst is enough for a reaction in bulk.
2. There may be some physical changes, but the catalyst remains unchanged in mass and chemical composition in a chemical reaction.
3. A catalyst itself cannot initiate a reaction. It means it can not start a reaction which is not taking place. But, if the reaction is taking place in a slow rate it can increase its rate.
4. A solid catalyst will be more effective if it is taken in a finely divided form.
5. A catalyst can catalyse a particular type of reaction, hence they are said to be specific in nature.
6. In an equilibrium reaction, presence of catalyst reduces the time for attainment of equilibrium and hence it does not affect the position of equilibrium and the value of equilibrium constant.
7. A catalyst is highly effective at a particular temperature called as optimum temperature.
8. Presence of a catalyst generally does not change the nature of products

   For example. $2SO_2 + O_2 \rightarrow 2SO_3$

   This reaction is slow in the absence of a catalyst, but fast in the presence of Pt catalyst

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